

1. DATA

Real-IAD

30 products, C1 camera
~150K images + masks

Gemini-2.5-Flash

trace generation
(img + red-mask + ref.)

AnomalyThink

6K SFT · 4.2K GRPO
· 4.2K held-out
stratified C1 (no
second-model filter)

2. SUPERVISED FINE-TUNING

SFT four-factor grid

3B/7B · frozen/unfrozen · 6K/15K
best: 7B-frozen-6K = **80.16%**

Teacher ablation

→ **Arm C (STaR)**
rollout ckpt-530 → Gemini judge:
keep / correct / rewrite
→ SFT *fresh base*
82.80% / 72.07% =
best model in thesis

3. GRPO (and its limit)

GRPO

4-component reward
(fmt + acc + type + loc), $G=4$

on the 6K-SFT init (80.16%):
→ **82.73 / 70.39** (lift)

on *strong* Arm-C init: **no gain**, ~2 pp
below init while reward rises —
holds for all 3 advantage estimators

4. EVALUATION

Evaluation (MMAD)

DS-MVTEC · VisA
+ held-out Real-IAD

Best: Arm-C SFT

82.80% / 72.07%
vs. IAD-R1 (our harness)
81.92% / 71.34%